 Development Operations

Kubernetes Basic Setup Minikube

 E41230292 Achmad Aryo Sangkelad

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Install Minikube

```
PS C:\Users\Sangkelad> winget install Kubernetes.minikube
The `msstore` source requires that you view the following agreements before using.
Terms of Transaction: https://aka.ms/microsoft-store-terms-of-transaction
The source requires the current machine's 2-letter geographic region to be sent to the
backend service to function properly (ex. "US").

Do you agree to all the source agreements terms?
[Y] Yes  [N] No: y
Found Kubernetes - Minikube - A Local Kubernetes Development Environment [Kubernetes.mi
nikube] Version 1.38.1
This application is licensed to you by its owner.
Microsoft is not responsible for, nor does it grant any licenses to, third-party packag
es.
Downloading https://github.com/kubernetes/minikube/releases/download/v1.38.1/minikube-i
nstaller.exe
  ██████████████████████████████  48.5 MB / 48.5 MB
Successfully verified installer hash
Starting package install...
Successfully installed
```

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Start Minikube

Windows PowerShell

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindo
ws

PS C:\Users\Sangkelad> minikube start --driver=docker
* minikube v1.38.1 on Microsoft Windows 11 Home Single Language 25H2
* Using the docker driver based on user configuration
! Starting v1.39.0, minikube will default to "containerd" container runtime. See #21973
 for more info.
* Using Docker Desktop driver with root privileges
* Starting "minikube" primary control-plane node in "minikube" cluster
* Pulling base image v0.0.50 ...
* Downloading Kubernetes v1.35.1 preload ...
    > preloaded-images-k8s-v18-v1...:  30.97 MiB / 272.45 MiB  11.37% 52.53 KiB
    > gcr.io/k8s-minikube/kicbase...:  32.33 MiB / 519.58 MiB  6.22% 55.10 KiB
    > gcr.io/k8s-minikube/kicbase...:  1.60 KiB / 519.58 MiB  0.00% 59 B p/s 28
    > index.docker.io/kicbase/sta...:  484.00 MiB / 519.58 MiB  93.15% 284.55 K
```

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Cek Nodes

```
PS C:\Users\Sangkelad> kubectl get nodes
NAME       STATUS   ROLES           AGE   VERSION
minikube   Ready    control-plane   82s   v1.35.1
PS C:\Users\Sangkelad>
```

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Buat Struktur Project

```
PS C:\Users\Sangkelad> mkdir kubernetes


    Directory: C:\Users\Sangkelad


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----         4/29/2026   9:18 AM                kubernetes


PS C:\Users\Sangkelad> cd kubernetes
PS C:\Users\Sangkelad\kubernetes> code .
PS C:\Users\Sangkelad\kubernetes>
```

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Membuat File Deployment & Service

Help kubernetes

Welcome deployment.yaml

deployment.yaml

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: web-nginx
spec:
  replicas: 3
  selector:
    matchLabels:
      app: web-nginx
  template:
    metadata:
      labels:
        app: web-nginx
    spec:
      containers:
      - name: nginx
        image: nginx
        ports:
        - containerPort: 80
```

Help kubernetes

Welcome deployment.yaml service.yaml

service.yaml

```yaml
apiVersion: v1
kind: Service
metadata:
  name: web-nginx-service
spec:
  selector:
    app: web-nginx
  ports:
    - port: 80
      targetPort: 80
  type: NodePort
```

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Menjalankan Yaml

```
PS C:\Users\Sangkelad\kubernetes> kubectl apply -f deployment.yaml
deployment.apps/web-nginx unchanged
PS C:\Users\Sangkelad\kubernetes> kubectl apply -f service.yaml
service/web-nginx-service created
PS C:\Users\Sangkelad\kubernetes>
```

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Lihat Semua Resource dan Cek Pods

```
PS C:\Users\Sangkelad\kubernetes> kubectl get all
NAME                              READY   STATUS              RESTARTS   AGE
pod/web-nginx-748d764f7-glbcv     0/1     ContainerCreating   0          82s
pod/web-nginx-748d764f7-s56hz     0/1     ContainerCreating   0          82s
pod/web-nginx-748d764f7-xv77p     0/1     ContainerCreating   0          82s

NAME                          TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)        AGE
service/kubernetes            ClusterIP   10.96.0.1        <none>        443/TCP        5m
service/web-nginx-service     NodePort    10.108.146.143   <none>        80:30692/TCP   20s

NAME                          READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/web-nginx     0/3     3            0           82s

NAME                                     DESIRED   CURRENT   READY   AGE
replicaset.apps/web-nginx-748d764f7      3         3         0       82s
PS C:\Users\Sangkelad\kubernetes>
```

```
PS C:\Users\Sangkelad\kubernetes> kubectl get pods
NAME                          READY   STATUS    RESTARTS   AGE
web-nginx-748d764f7-glbcv     1/1     Running   0          118s
web-nginx-748d764f7-s56hz     1/1     Running   0          118s
web-nginx-748d764f7-xv77p     1/1     Running   0          118s
PS C:\Users\Sangkelad\kubernetes>
```

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Monitoring Live

```
PS C:\Users\Sangkelad\kubernetes> kubectl get pods -w
NAME                          READY   STATUS    RESTARTS   AGE
web-nginx-748d764f7-glbcv     1/1     Running   0          2m10s
web-nginx-748d764f7-s56hz     1/1     Running   0          2m10s
web-nginx-748d764f7-xv77p     1/1     Running   0          2m10s
```

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Membuka Aplikasi

```
PS C:\Users\Sangkelad\kubernetes> minikube service web-nginx-service
```

NAMESPACE	NAME	TARGET PORT	URL
default	web-nginx-service	80	http://192.168.49.2:30692

```
🔗 Starting tunnel for service web-nginx-service.
```

NAMESPACE	NAME	TARGET PORT	URL
default	web-nginx-service		http://127.0.0.1:59344

```
🎉 Opening service default/web-nginx-service in default browser...
❗ Because you are using a Docker driver on windows, the terminal needs to be open to run it.
```

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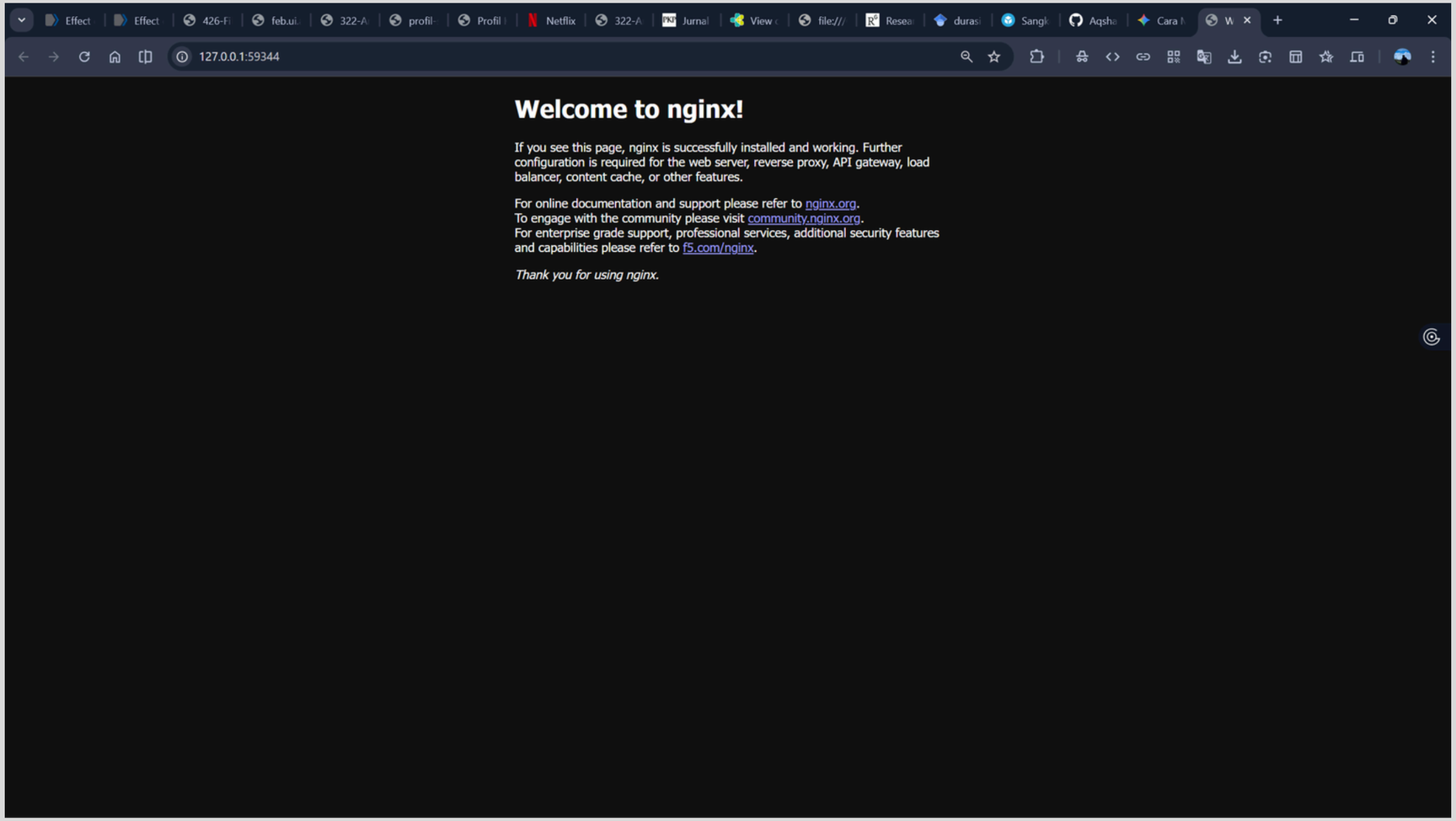
Membuka Aplikasi



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Scaling Pod & Lihat Deployment Live

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
          1s
web-nginx-748d764f7-hthst   0/1     ContainerCreating   0
 1s
web-nginx-748d764f7-z4779   0/1     ContainerCreating   0
 1s
web-nginx-748d764f7-z4779   1/1     Running             0
 4s
web-nginx-748d764f7-hthst   1/1     Running             0
 6s
web-nginx-748d764f7-hthst   1/1     Terminating         0
 75s
web-nginx-748d764f7-z4779   1/1     Terminating         0
 75s
web-nginx-748d764f7-hthst   1/1     Terminating         0
 75s
web-nginx-748d764f7-z4779   1/1     Terminating         0
 75s
web-nginx-748d764f7-hthst   0/1     Completed           0
 75s
web-nginx-748d764f7-z4779   0/1     Completed           0
 75s
web-nginx-748d764f7-hthst   0/1     Completed           0
 75s
web-nginx-748d764f7-hthst   0/1     Completed           0
 75s
web-nginx-748d764f7-z4779   0/1     Completed           0
 75s
web-nginx-748d764f7-z4779   0/1     Completed           0
 75s
```

```
PS C:\Users\Sangkelad\kubernetes> kubectl scale deployment web-ngi
nx --replicas=5ngkelad\kubernetes>  kubectl scale deployment web-n
ginx --replicas=3ps/web-nginx scaled
deployment.apps/web-nginx scaledtes>
PS C:\Users\Sangkelad\kubernetes> minikube node add
```

```
PS C:\Users\Sangkelad\kubernetes> kubectl get deployment -w
NAME        READY   UP-TO-DATE   AVAILABLE   AGE
web-nginx   3/3     3            3           7m37s
```

kubectl

kub...
pow...
kub...

0 0 Ln 5, Col 6 Spaces: 2 UTF-8 CRLF { } YAML Go Live Prettier

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Menambah Worker Node

```
PS C:\Users\Sangkelad\kubernetes> minikube node add
😄  Adding node m02 to cluster minikube as [worker]
❗  Cluster was created without any CNI, adding a node to it might cause broken networking.
👍  Starting "minikube-m02" worker node in "minikube" cluster
🚜  Pulling base image v0.0.50 ...
💾  Downloading Kubernetes v1.35.1 preload ...
    > preloaded-images-k8s-v18-v1...:  272.45 MiB / 272.45 MiB  100.00% 4.44 Mi
🔥  Creating docker container (CPUs=2, Memory=2200MB) ...
🐳  Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
🔎  Verifying Kubernetes components...
🏄  Successfully added m02 to minikube!
PS C:\Users\Sangkelad\kubernetes> kubectl get nodes
NAME           STATUS     ROLES           AGE   VERSION
minikube       Ready      control-plane   18m   v1.35.1
minikube-m02   NotReady   <none>          9s    v1.35.1
PS C:\Users\Sangkelad\kubernetes> minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured

minikube-m02
type: Worker
host: Running
kubelet: Running

PS C:\Users\Sangkelad\kubernetes>
```

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Menghapus Project

```
PS C:\Users\Sangkelad\kubernetes> kubectl delete -f deployment.yaml
>> kubectl delete -f service.yaml
deployment.apps "web-nginx" deleted from default namespace
service "web-nginx-service" deleted from default namespace
PS C:\Users\Sangkelad\kubernetes>
```

THANK YOU